

SIMATS
ENGINEERING

Secure Exit

SIMATS Online Programming
Lab

JAVA PROGRAMMING

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Multiple Inheritance

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QUESTIONS

Q1

Food Delivery Billing System
(Multiple Inheritance using Two
Interfaces)
10 marks

Q2

Hotel Billing System (Multiple
Inheritance using One Class
and One Interface)
10 marks

2/2 answered

Q1 Food Delivery Billing System (Multiple Inheritance using Two Interfaces) 10 marks

A food delivery application calculates the final bill for a customer's order by considering the food cost, delivery charges, and coupon discounts.

Create two interfaces:

DeliveryCharge with a method calculateDeliveryCharge().

CouponDiscount with a method calculateDiscount().

Create a class FoodOrder that implements both interfaces.

The delivery charge is calculated as follows:

If the food cost is ₹500 or more, the delivery charge is ₹0.

Otherwise, the delivery charge is ₹50.

The coupon discount is calculated as follows:

If the coupon code is SAVE10, the customer receives a 10% discount on the food cost.

Otherwise, no discount is applied.

The final bill is calculated using the formula:

Final Bill = Food Cost + Delivery Charge – Discount

Display the food cost, delivery charge, discount amount, and final bill.

Input Format

Food Cost

Coupon Code

Output Format

Food Cost : <food_cost>

Delivery Charge : <delivery_charge>

Discount : <discount>

Final Bill : <final_bill>

Sample Input

600

SAVE10

Sample Output

Food Cost : 600.0

Delivery Charge : 0.0

Discount : 60.0

Final Bill : 540.0

Your Code

```
1 import java.util.*;
2 interface DeliveryCharge{
3     double calculateDeliveryCharge(double f);
4 }
5 interface CouponDiscount{
6     double calculateCouponDiscount(double f,String c);
7 }
8 public class FoodOrder implements DeliveryCharge,CouponDiscount{
```

```
9      public double calculateDeliveryCharge(double f) {
10          return f>=500?0:50;
11      }
12      public double calculateCouponDiscount(double f,String c){
13          return c.equals("SAVE10")?f*0.1:0;
14      }
15      public static void main(String[] args){
16          Scanner sc=new Scanner(System.in);
17          double f=sc.nextDouble();
18          String c=sc.next();
19          FoodOrder o=new FoodOrder();
20          double d=o.calculateDeliveryCharge(f);
21          double dis=o.calculateCouponDiscount(f,c);
22          System.out.println("Food Cost : "+f);
23          System.out.println("Delivery Charge : "+d);
24          System.out.println("Discount : "+dis);
25          System.out.println("Final Bill : "+(f+d-dis));
26      }
27 }
```

Input

```
600
SAVE10
```

Output

```
Food Cost : 600.0  
Delivery Charge : 0.0  
Discount : 60.0  
Final Bill : 540.0
```

Visible Test Cases

Test Case 1



Test Case 2



Test Case 3



10.00 / 10 marks

 Submitted

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2/2 answered

Q2

Hotel Billing System (Multiple Inheritance using One Class and One Interface)

10
marks

Problem Statement

A hotel wants to generate the final bill for its guests by adding a service charge to the room rent.

Create a class Room that stores the following details:

Room Number

Guest Name

Room Rent

Create an interface ServiceCharge with a method:

```
double calculateServiceCharge();
```

Create a class HotelBill that extends the Room class and implements the ServiceCharge interface.

The service charge is calculated as follows:

If the room rent is ₹5000 or more, the service charge is 12% of the room rent.
Otherwise, the service charge is 8% of the room rent.

The final bill is calculated as:

Final Bill = Room Rent + Service Charge

Display the room details, room rent, service charge, and final bill.

Input Format

Room Number

Guest Name

Room Rent

Output Format

Room Number : <room_number>

Guest Name : <guest_name>

Room Rent : <room_rent>

Service Charge : <service_charge>

Final Bill : <final_bill>

Sample Input

205

Ananya

6000

Sample Output

Room Number : 205

Guest Name : Ananya

Room Rent : 6000.0

Service Charge : 720.0

Final Bill : 6720.0

Your Code

```
1 import java.util.*;
2 class Room{
3     int r;
4     String n;
5     double rent;
6     public static void main(String[] args){
7         Scanner sc=new Scanner(System.in);
8         HotelBill h = new HotelBill();
9         h.r=sc.nextInt();
10        sc.nextLine();
11        h.n=sc.nextLine();
12        h.rent=sc.nextDouble();
13        double s=h.calculateServiceCharge();
14        System.out.println("Room Number : "+h.r);
15        System.out.println("Guest Name : "+h.n);
16        System.out.println("Room Rent : "+h.rent);
17        System.out.println("Service Charge : "+s);
18        System.out.println("Final Bill : "+(h.rent+s));
19    }
20 }
21 interface ServiceCharge {
22     double calculateServiceCharge();
23 }
24 class HotelBill extends Room implements ServiceCharge{
25     public double calculateServiceCharge(){
26         return rent>=5000?rent*0.12:rent*0.08;
27     }
28 }
```

Input

241

sam

2345



Output

```
Room Number : 241
Guest Name : sam
Room Rent : 2345.0
Service Charge : 187.6
Final Bill : 2532.6
```



Visible Test Cases

Test Case 1



Test Case 2



Test Case 3



10.00 / 10 marks

Submitted